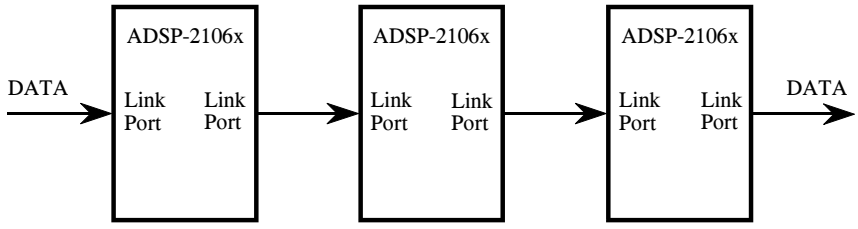
a. Data flow multiprocessing



b. Cluster multiprocessing

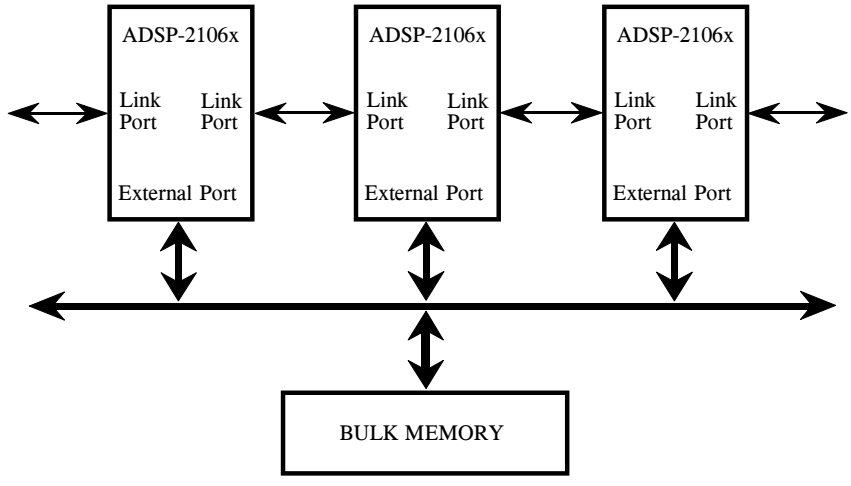


FIGURE 28-12
Multiprocessing configurations. Multiprocessor systems typically use one of two schemes to communicate between processor nodes, (a) dedicated point-to-point communication channels, or (b) a shared global memory accessed over a parallel bus.

strategy. In (b), the processors interact through a single shared global memory, accessed over a parallel bus (i.e., the external port). Figure 28-13 shows another way that a large number of processors can be combined into a single system, a 2D or 3D "mesh." Each of these configuration will have relative advantages and disadvantages for a particular task.

To make the programmer's life easier, the SHARC family uses a *unified address space*. This means that the 4 Gigaword address space, accessed by the 32 bit address bus, is divided among the various processors that are working together. To transfer data from one processor to another, simply read from or write to the appropriate memory locations. The SHARC internal logic takes care of the rest, transferring the data between processors at a rate as high as 240 Mbytes/sec (at 40 MHz).